

Longitudinal Microbiome Profile in Allogeneic Hematopoietic Stem Cell Transplantation: Results of a Prospective Study in 100 Patients

Raffaella Greco, MD,^{*1} Nicasio Mancini, MD,^{*2} Renée Pasciuta,^{*2} Maria Chiara Barbanti, MD,^{*1}
Mara Morelli, MD,^{*1} Luca Vago, MD PhD,^{1,3} Fabio Giglio, MD,^{*1} Maria Teresa Lupo
Stanghellini, MD,^{*1} Alessandra Forcina, MD,^{*1} Laura Infurnari,^{*2} Sarah Marktel, MD,^{*1}
Andrea Angelo Assanelli, MD,^{*1} Matteo Giovanni Carrabba, MD,^{*1} Massimo Bernardi, MD,^{*4}
Consuelo Corti, MD,^{*1} Jacopo Peccatori, MD,^{*1} Massimo Clementi, MD,^{*5} Fabio Ciceri, MD^{*6,1}

¹Hematology and Bone Marrow Transplantation Unit, IRCCS San Raffaele Scientific Institute, Milan, Italy

²Laboratory of Microbiology and Virology, Vita-Salute San
Raffaele University, San Raffaele Scientific Institute, Milan, Italy

³Unit of Immunogenetics, Leukemia Genomics and Immunobiology,
IRCCS San Raffaele Scientific Institute, Milan, Italy

⁴Hematology and Bone Marrow Transplantation Unit, IRCCS San Raffaele Scientific Institute, Milano, Italy

⁵Laboratory of Microbiology and Virology, Vita-Salute San Raffaele
University, IRCCS San Raffaele Scientific Institute, Milan, Italy

⁶San Raffaele University Hospital and Scientific Institute, Milan, Italy

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Abstract

BACKGROUND: Allogeneic HSCT (allo-HSCT) is the only cure for several patients with hematological malignancies. Recent advances in therapies have significantly reduced transplant-related mortality (TRM); nevertheless infections and graft-versus-host disease (GvHD) still represent major complications. Recent studies indicate that patients undergo dramatic alterations of intestinal microbiota during allo-HSCT, potentially affecting the outcome. An alarming emergence of gram-negative multi-drug-resistant (GN-MDR) pathogens has been increasingly reported from various cancer centers, and the primary site of colonization is mainly the gut. Moreover, GvHD was associated with major shifts in the composition of intestinal microbiota, able to modulate the severity of intestinal inflammation.

METHODS: Between October 2014 and March 2016, we have conducted a prospective observational study to examine the intestinal microbiota by NGS techniques in 100 consecutive adult patients, who received allo-HSCT for high-risk hematological malignancies (63.5% acute leukemia). Stem cell donors were family haploidentical (n=45), HLA identical sibling (n=15), unrelated volunteer (n=35), cord blood (n=5). Stem cell source was mainly T-cell replete PBSCs. Fecal specimens have been collected before conditioning (T0), during aplasia (T10) and after engraftment (T30). The fecal microbiome have been analyzed using the 454 GS Junior System, and QIIME software. We defined a threshold of normality for the main phyla on the basis of the control.

RESULTS:

We observed an important difference in relative percentages of phyla populating the gut between our pts and control. Our patients showed a progressive reduction of the intestinal microbial diversity (alpha diversity) during the transplant process, with a specific decrease of anaerobic species belonging to both Bacteroidetes and Firmicutes phyla, and a relative increase of facultative anaerobic gram-positive families such as Enterococcaceae and Staphylococcaceae. A great variation was observed, particularly between T0 and T30, where we found a significant decrease in alpha diversity ($p < 0.001$).

A significantly different distribution in the baseline microbiome was reported in patients who will experience different clinical outcomes. Patients (n=23) who developed a microbiologically-confirmed sepsis show an increase of Proteobacteria phylum (cut-off 5%; $p < 0.01$, RR= 5.38), and a decrease in Lachnospiraceae (cut-off 10%; $p < 0.02$ and RR=2.77). Sepsis by GN-MDR bacteria (n=15) was strongly associated to an increase of Proteobacteria phylum ($p < 0.01$ and RR=5.38 at T0). Microbiome changes preceded the development of severe sepsis and septic shock in 21 patients (increase of Proteobacteria, $p < 0.01$ and RR= 2.43), underlying its potential for the prediction of these clinically relevant conditions and its impact on overall survival (increase of Proteobacteria $p < 0.01$ and RR= 2.69; decrease in Lachnospiraceae, $p < 0.01$ and RR=2.83).

We identified by multivariate analysis some associations between the relative increase of a specific phylum and clinical variables at baseline. Previous allo-HSCT and prior infections represented strong risk factors for developing colonization by Proteobacteria (exceeding 5%). Sepsis by GN-MDR bacteria was associated to increase in Enterobacteriaceae(exceeding 5%; $p < 0.011$).

Interestingly, significant microbioma changes were observed at 10 days after transplant, in patients who developed a subsequent acute GvHD, with a predominant role played by gram-positive bacteria belonging to Firmicutes. More in details, the presence of Lachnospiraceae was associated to a decreased

risk of developing acute GvHD ($p=0.04$ and $RR= 4.35$), whereas dominance of Enterococcaceae ($p<0.01$ and $RR=3.23$) and Staphylococcaceae ($p<0.01$ and $RR=3.5$) was associated to its increased incidence.

CONCLUSIONS:

Longitudinal study of microbioma profile allows early identification of patients at risk for major transplant-related complications such as sepsis by GN-MDR or acute GvHD, offering a new tool for individualized pre-emptive or therapeutical strategies to improve the outcome of allo-HSCT.

Disclosures

Ciceri: *MolMed SpA*: Consultancy.

Author notes

*Asterisk with author names denotes non-ASH members.